StayWise: Hotel Booking Analytics for Revenue & Cancellation Optimisation

Newton School of Technology · Section B, Group G-1

Team

- Sarthak Mishra (Project Lead)
- Ganga Raghuwanshi (Data Lead)
- Musthyala Sadhvik (ETL Lead)
- Kunal Dev Sahu (Analysis Lead)
- Soumen Dass (Visualisation Lead)
- Deepak Mishra (Strategy Lead)
- Lakshay Yadav (PPT & Quality Lead)

Resources

- **GitHub:** SectionB_G-1_StayWise
- **Tableau Executive:** StayWise Dashboard 1
- **Tableau Operational:** StayWise Dashboard 2

The Revenue Leakage Problem

Industry Context

Frictionless OTA cancellation policies disrupt hotel revenue, staffing, and dynamic pricing — leaving General Managers and Revenue Managers scrambling to fill last-minute vacancies.

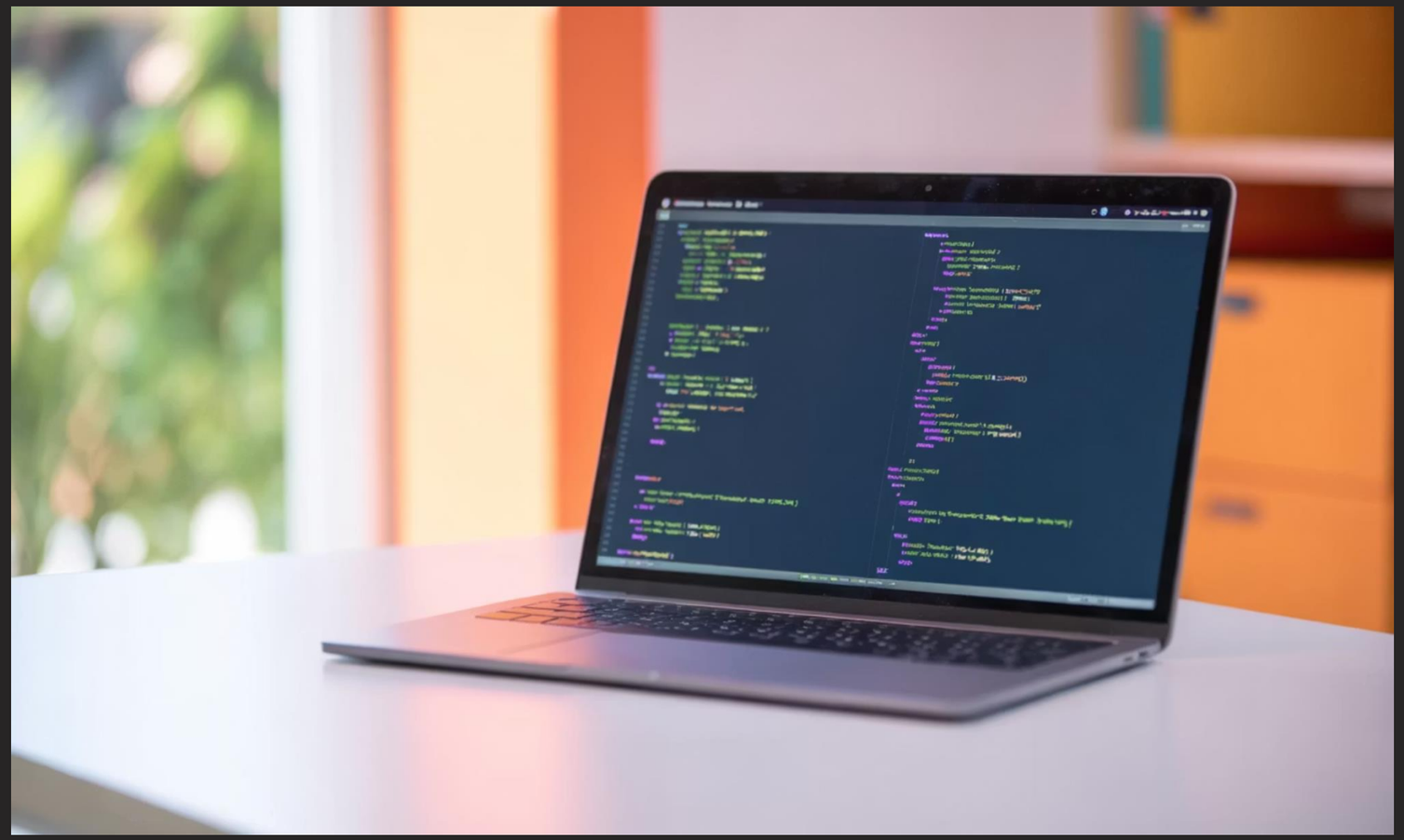
Core Question

What are the primary behavioural and transactional drivers of hotel booking cancellations, and how can policies be optimised to mitigate this risk?

Objective

Identify high-risk booking profiles to deploy targeted deposit policies, direct-channel incentives, and dynamic pricing that maximise net revenue.

Data Engineering: Source to Pipeline



Dataset Overview

Source: Hotel Booking Demand Dataset (Kaggle) · July 2015 – August 2017
Raw: 119,209 rows · 49 columns → **Processed:** 87,219 rows · 29 columns

ETL Steps (Python)

- **Null Handling:** Dropped heavily null `company` column; imputed `country`, `agent`, `children`
- **Outlier Treatment:** Capped ADR at £5,000 to remove erroneous billing inputs
- **Feature Engineering:** Created `total_nights`, `total_revenue`, `clv_proxy`, `is_peak_season`

Code committed in `scripts/etl_pipeline.py` and `notebooks/02_cleaning.ipynb`

KPI & Metrics Framework

Four KPIs were engineered to directly track revenue leakage and recovery opportunity, computed in `notebooks/03_eda.ipynb`.

Cancellation Rate %

Direct measure of revenue leakage.

`sum(is_canceled) / count(bookings)`

Total Revenue

Gross financial value underpinning dynamic pricing.

`total_nights × ADR`

Revenue per Guest

Measures demographic profitability.

`total_revenue / total_guests`

CLV Proxy

Realised economic value per booking.

`total_revenue × (1 – is_canceled)`

Key Insights from Exploratory Analysis

1

City vs Resort Hotels

City Hotels suffer a **41% cancellation rate** vs only 27% for Resort Hotels — far more volatile and policy-sensitive.

2

Phantom Demand

Bookings made **more than 180 days in advance** cancel at a disproportionately high rate, inflating the demand picture.

3

No-Deposit Risk

"No Deposit" bookings account for nearly all cancellations. Non-Refundable deposits practically eliminate cancellation risk.

4

OTA Channel Risk

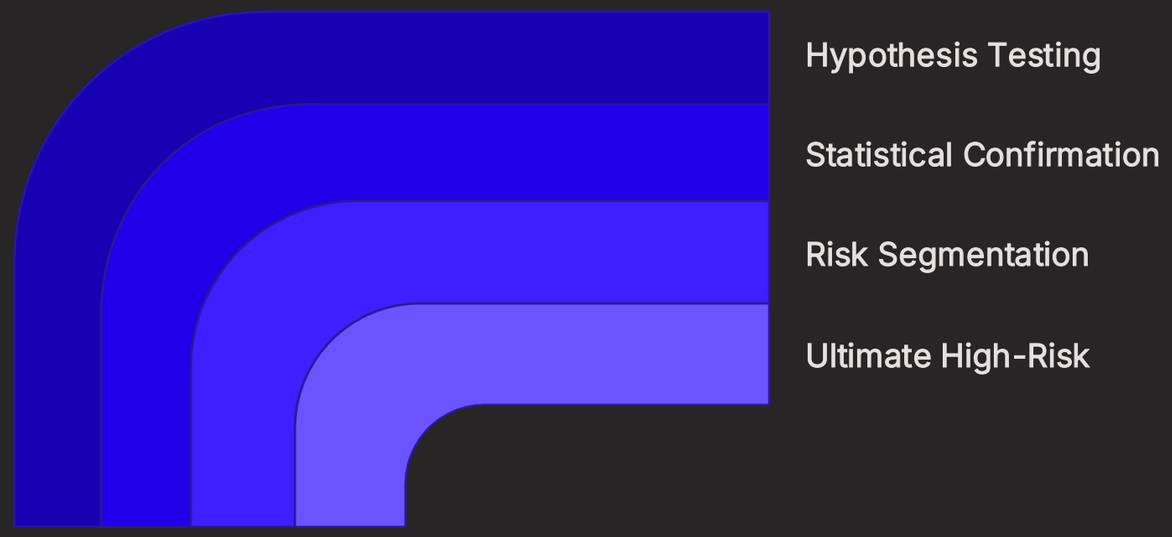
TA/TO (OTA) channels cancel **significantly more** than direct hotel website bookings, amplifying commission costs.

5

Summer Peak Exposure

July–August drives the **highest volume and ADR** — making cancellations here the most financially damaging of the year.

Advanced Analysis: Confirming the Risk Profile



Statistical Confirmation

An independent samples **t-test** (scipy.stats) confirmed with **$p < 0.001$** that cancelled bookings carry a significantly higher mean lead time than fulfilled bookings.

The "Ultimate High-Risk" Profile

⚠ **City Hotel + No Deposit + TA/TO Channel + Lead Time > 180 days** — a laser-focused target for policy intervention.

Libraries used: pandas, scipy, seaborn, matplotlib

Tableau Dashboard Walkthrough

Executive View

High-level KPI summary (Total Revenue, Cancel Rate, Bookings) and monthly trend lines — designed for the GM to assess property health at a glance.

Open Executive Dashboard →

Operational View

Drill-down charts by lead time, deposit type, and market segment — built for the Revenue Analyst adjusting daily pricing tactics.

Open Operational Dashboard →

Interactive Feature

A **Market Segment filter** dynamically updates the Risk Profile Matrix to isolate OTA dependency — enabling live scenario analysis.

Strategic Recommendations

1. Targeted Deposit Policies

Mandate **non-refundable deposits** for all bookings made more than 180 days out. Converts speculative placeholders into committed, secured revenue.

2. "Book Direct" Loyalty Programme

Offer exclusive perks (breakfast, parking) for direct web or phone bookings. Reduces the **15–25% OTA commission** fee and cuts cancellation rates simultaneously.

3. Dynamic Peak Pricing

Deploy stricter cancellation windows and rate premiums during July–August. Protects the **highest-value inventory** from last-minute revenue loss.

Business Impact & Value at Stake

10%

Conversion Target

Converting just 10% of high-risk "No Deposit" long-lead bookings to Non-Refundable secures massive top-line room revenue.

5%

Channel Shift

Shifting just 5% of bookings from OTAs to direct channels drops 15–25% commission savings directly to the bottom line.

Cost of Inaction: Ignoring OTA dominance and speculative booking behaviour guarantees continued RevPAR erosion, unpredictable staffing, and avoidable commission expenditure year-on-year.

Limitations & Next Steps

Current Limitations

Pre-Pandemic Data

Dataset covers 2015–2017 and misses modern "bleisure" and remote-work travel patterns.

Missing Signals

No competitor pricing, local event calendars, or macro-economic demand indicators are included.

Future Roadmap

ML Prediction Engine

Build a **Random Forest / XGBoost** classifier to assign a real-time "Cancellation Probability Score" at checkout, triggering dynamic deposit rules automatically.

Live PMS Integration

Connect Tableau dashboards to a **live Property Management System API** for real-time revenue tracking and operational alerts.

Team & Contribution

Dataset & Sourcing

Ganga Raghuwanshi

ETL & Cleaning

Musthyala Sadhvik

EDA & Analysis

Kunal Dev Sahu

KPI Analysis

Deepak Mishra & Soumen Dass

Tableau Dashboard

Soumen Dass

Report Writing

Sarthak Mishra

Business Recommendation

Deepak Mishra

PPT & Viva Prep

Sarthak Mishra & Lakshay Yadav